

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 1000.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 432

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 27

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 22512162891.258385

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 222263.08453839074

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 147787.1156539602

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.00087898503

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.4999999997657625

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 1.1303901049932525

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition -187615.24172287967

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 0.0141

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 1.1660

Parameter bounds [0.000E+000; 3.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 0.1104

Parameter bounds [0.000E+000; 3.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.0000

Parameter bounds [0.000E+000; 1.000E+000]

Diffusion-flame standoff pressure-factor C_{rxn} 50

Parameter bounds [0.000E+000; 5.000E+001]

Diffusion-flame standoff size-exponent m 0.219675

Parameter bounds [0.000E+000; 2.000E+000]

Bimodal-packing heat-feedback K_{pack} 10

Parameter bounds [0.000E+000; 1.000E+001]
Diffusion-flame standoff pressure-exponent n_p 2.11429
Parameter bounds [2.000E+000; 4.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999981723701.445
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 184974.40199247756
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 253704006643.3498
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 167233.22029012255
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.00010243783812710663
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 1.362476711039231E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 2.8947610497399888E-11
Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 19050163.68838125
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 86415.44416013312
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 6189135.648419971
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 50002.75523573763
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.9412415282324234
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 4.449114241412574E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 5.943514180780209E-10
Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 7152404613979.619
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 167877.9291099764
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 213239956.75508934
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 73676.57689886919
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.0022428726403525
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 2.684331945057925E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 5.560418632724781E-06
Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.0231
Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s

Calculated burning rate 15.48 mm/s

First Conditional Fuel

Surface temperature 757.15 K

Linear burning rate 11.3 mm/s

Mass decomposition rate $18.75 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $9.4\text{e}+06 \text{ W/m}^2$

Kinetic flame height 31.4 μm

Average kinetic flame density 1.78 kg/m^3

Average kinetic flame temperature 2,232.07 K

Binder sublimation heat flux density $9.4\text{e}+06 \text{ W/m}^2$

Surface heat balance error -0.0028 W/m^2

Secondary Conditional Fuel

Surface temperature 748.53 K

Linear burning rate 8.31 mm/s

Mass decomposition rate $13.79 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $104,815.24 \text{ W/m}^2$

Kinetic flame height 1,542.21 μm

Average kinetic flame density 2.55 kg/m^3

Average kinetic flame temperature 1,556.76 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.27\text{e}+06 \text{ W/m}^2$

Kinetic flame height 37.06 μm

Average kinetic flame density 3.39 kg/m^3

Average kinetic flame temperature 1,169.76 K

Heat flux density due to metal combustion, $3.23\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.42 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.015 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$

Skeleton layer thickness $7.15\text{e-}05 \text{ m}$

Pore diameter $9.44\text{e-}06 \text{ m}$

Diffusion flame heat flux density $13,605.98 \text{ W/m}^2$

Diffusion flame height 20,986.91 μm

Heat flux density to surface within skeletal structure $1.97\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $67,193.64 \text{ W/m}^2$

Total heat flux density to surface $6.73\text{e}+06 \text{ W/m}^2$

Binder sublimation heat flux density $6.73\text{e}+06 \text{ W/m}^2$

Surface heat balance error $6.67\text{e-}04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s

Calculated burning rate 35.12 mm/s

First Conditional Fuel

Surface temperature 809.45 K

Linear burning rate 63.32 mm/s

Mass decomposition rate $105.04 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $6.09\text{e}+07 \text{ W/m}^2$

Kinetic flame height 4.76 μm

Average kinetic flame density 7.13 kg/m^3

Average kinetic flame temperature 2,258.23 K

Binder sublimation heat flux density $6.09\text{e}+07 \text{ W/m}^2$

Surface heat balance error -0.0028 W/m^2

Secondary Conditional Fuel

Surface temperature 766.93 K

Linear burning rate 15.88 mm/s

Mass decomposition rate $26.35 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 268,278.25 W/m²

Kinetic flame height 595.68 μm

Average kinetic flame density 10.28 kg/m³

Average kinetic flame temperature 1,565.97 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.98e+06 W/m²

Kinetic flame height 16.53 μm

Average kinetic flame density 13.65 kg/m³

Average kinetic flame temperature 1,178.97 K

Heat flux density due to metal combustion, 6.62e+06 W/m²

Effective thermal conductivity 0.42 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 0.014 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 3.36e-05 m

Pore diameter 8.79e-06 m

Diffusion flame heat flux density 135,646.4 W/m²

Diffusion flame height 2,091.52 μm

Heat flux density to surface within skeletal structure 3.83e+06 W/m²

Heat flux density to surface outside skeletal structure 179,653.53 W/m²

Total heat flux density to surface 1.36e+07 W/m²

Binder sublimation heat flux density 1.36e+07 W/m²

Surface heat balance error -0.0014 W/m²

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s

Calculated burning rate 40.24 mm/s

First Conditional Fuel

Surface temperature 828.78 K

Linear burning rate 113.29 mm/s

Mass decomposition rate 187.94 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.14e+08 W/m²

Kinetic flame height 2.52 μm

Average kinetic flame density 11.38 kg/m³

Average kinetic flame temperature 2,267.89 K

Binder sublimation heat flux density 1.14e+08 W/m²

Surface heat balance error 0.014 W/m²

Secondary Conditional Fuel

Surface temperature 769.8 K

Linear burning rate 17.52 mm/s

Mass decomposition rate 29.06 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 414,837.37 W/m²

Kinetic flame height 384.54 μm

Average kinetic flame density 16.46 kg/m³

Average kinetic flame temperature 1,567.4 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 7.28e+06 W/m²

Kinetic flame height 11.28 μm

Average kinetic flame density 21.86 kg/m³

Average kinetic flame temperature 1,180.4 K

Heat flux density due to metal combustion, 7.38e+06 W/m²

Effective thermal conductivity 0.42 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 0.014 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 3e-05 m

Pore diameter 8.69e-06 m

Diffusion flame heat flux density 329,058 W/m²
Diffusion flame height 861.31 μm
Heat flux density to surface within skeletal structure 3.98e+06 W/m²
Heat flux density to surface outside skeletal structure 302,134.46 W/m²
Total heat flux density to surface 1.51e+07 W/m²
Binder sublimation heat flux density 1.51e+07 W/m²
Surface heat balance error 4.96e-04 W/m²

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
Calculated burning rate 15.09 mm/s

First Conditional Fuel

Surface temperature 757.15 K
Linear burning rate 11.3 mm/s
Mass decomposition rate 18.75 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 9.4e+06 W/m²
Kinetic flame height 31.4 μm
Average kinetic flame density 1.78 kg/m³
Average kinetic flame temperature 2,232.07 K
Binder sublimation heat flux density 9.4e+06 W/m²
Surface heat balance error -0.0028 W/m²

Secondary Conditional Fuel

Surface temperature 757.15 K
Linear burning rate 11.3 mm/s
Mass decomposition rate 18.75 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 77,268.29 W/m²
Kinetic flame height 2,080.87 μm
Average kinetic flame density 2.54 kg/m³
Average kinetic flame temperature 1,561.07 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.64e+07 W/m²
Kinetic flame height 7.07 μm
Average kinetic flame density 2.35 kg/m³
Average kinetic flame temperature 1,690.07 K
Heat flux density due to metal combustion, 3.39e+06 W/m²
Effective thermal conductivity 0.31 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.09 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 5e-05 m
Pore diameter 9.12e-06 m
Diffusion flame heat flux density 26,382.67 W/m²
Diffusion flame height 11,055.95 μm
Heat flux density to surface within skeletal structure 9.32e+06 W/m²
Heat flux density to surface outside skeletal structure 53,090.23 W/m²
Total heat flux density to surface 9.4e+06 W/m²
Binder sublimation heat flux density 9.4e+06 W/m²
Surface heat balance error -0.0018 W/m²

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 36.66 mm/s

First Conditional Fuel

Surface temperature 809.45 K
Linear burning rate 63.32 mm/s
Mass decomposition rate 105.04 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface $6.09\text{e}+07 \text{ W/m}^2$
Kinetic flame height $4.76 \text{ }\mu\text{m}$
Average kinetic flame density 7.13 kg/m^3
Average kinetic flame temperature $2,258.23 \text{ K}$
Binder sublimation heat flux density $6.09\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0028 W/m^2

Secondary Conditional Fuel

Surface temperature 772.56 K
Linear burning rate 19.24 mm/s
Mass decomposition rate $31.92 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $221,759.83 \text{ W/m}^2$
Kinetic flame height $718.09 \text{ }\mu\text{m}$
Average kinetic flame density 10.26 kg/m^3
Average kinetic flame temperature $1,568.78 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.37\text{e}+07 \text{ W/m}^2$
Kinetic flame height $2.9 \text{ }\mu\text{m}$
Average kinetic flame density 9.48 kg/m^3
Average kinetic flame temperature $1,697.78 \text{ K}$
Heat flux density due to metal combustion, $6.02\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.31 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.085 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E-}08$
Skeleton layer thickness $2.69\text{e-}05 \text{ m}$
Pore diameter $8.6\text{e-}06 \text{ m}$

Diffusion flame heat flux density $295,480.18 \text{ W/m}^2$
Diffusion flame height $981.94 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $1.63\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $170,049.82 \text{ W/m}^2$
Total heat flux density to surface $1.67\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.67\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0036 W/m^2

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s
Calculated burning rate 40.76 mm/s

First Conditional Fuel

Surface temperature 828.78 K
Linear burning rate 113.29 mm/s
Mass decomposition rate $187.94 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.14\text{e}+08 \text{ W/m}^2$
Kinetic flame height $2.52 \text{ }\mu\text{m}$
Average kinetic flame density 11.38 kg/m^3
Average kinetic flame temperature $2,267.89 \text{ K}$
Binder sublimation heat flux density $1.14\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.014 W/m^2

Secondary Conditional Fuel

Surface temperature 773.24 K
Linear burning rate 19.69 mm/s
Mass decomposition rate $32.66 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $369,472.69 \text{ W/m}^2$
Kinetic flame height $430.82 \text{ }\mu\text{m}$
Average kinetic flame density 16.44 kg/m^3
Average kinetic flame temperature $1,569.12 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $1\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.85 μm
Average kinetic flame density 15.19 kg/m^3
Average kinetic flame temperature 1,698.12 K
Heat flux density due to metal combustion, 6.17e+06 W/m^2
Effective thermal conductivity 0.31 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.22 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 0.085 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 2.62e-05 m
Pore diameter 8.58e-06 m
Diffusion flame heat flux density 773,103.48 W/m^2
Diffusion flame height 375.21 μm
Heat flux density to surface within skeletal structure 1.61e+07 W/m^2
Heat flux density to surface outside skeletal structure 313,561.48 W/m^2
Total heat flux density to surface 1.72e+07 W/m^2
Binder sublimation heat flux density 1.72e+07 W/m^2
Surface heat balance error 0.0018 W/m^2

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
Calculated burning rate 10.79 mm/s

First Conditional Fuel

Surface temperature 757.15 K
Linear burning rate 11.3 mm/s
Mass decomposition rate 18.75 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 9.4e+06 W/m^2
Kinetic flame height 31.4 μm
Average kinetic flame density 1.78 kg/m^3
Average kinetic flame temperature 2,232.07 K
Binder sublimation heat flux density 9.4e+06 W/m^2
Surface heat balance error -0.0028 W/m^2

Secondary Conditional Fuel

Surface temperature 727.19 K
Linear burning rate 3.77 mm/s
Mass decomposition rate 6.25 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 4,532.47 W/m^2
Kinetic flame height 6,945.66 μm
Average kinetic flame density 4.49 kg/m^3
Average kinetic flame temperature 884.6 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 871,348.21 W/m^2
Kinetic flame height 58.97 μm
Average kinetic flame density 4.03 kg/m^3
Average kinetic flame temperature 984.1 K
Heat flux density due to metal combustion, 4.51e+06 W/m^2
Effective thermal conductivity 1.42 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 1.41 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 0.0063 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 1.8e-04 m
Pore diameter 1.03e-05 m

Diffusion flame heat flux density 4,508.11 W/m^2
Diffusion flame height 63,281.75 μm
Heat flux density to surface within skeletal structure 2.84e+06 W/m^2
Heat flux density to surface outside skeletal structure 2,135.84 W/m^2
Total heat flux density to surface 2.85e+06 W/m^2

Binder sublimation heat flux density $2.85\text{e}+06 \text{ W/m}^2$
Surface heat balance error $1.06\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 26.2 mm/s

First Conditional Fuel

Surface temperature 809.45 K
Linear burning rate 63.32 mm/s
Mass decomposition rate $105.04 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $6.09\text{e}+07 \text{ W/m}^2$
Kinetic flame height 4.76 μm
Average kinetic flame density 7.13 kg/m^3
Average kinetic flame temperature 2,258.23 K
Binder sublimation heat flux density $6.09\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0028 W/m^2

Secondary Conditional Fuel

Surface temperature 747.34 K
Linear burning rate 7.97 mm/s
Mass decomposition rate $13.21 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $10,410.06 \text{ W/m}^2$
Kinetic flame height 2,830.5 μm
Average kinetic flame density 17.99 kg/m^3
Average kinetic flame temperature 894.67 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.75\text{e}+06 \text{ W/m}^2$
Kinetic flame height 28.26 μm
Average kinetic flame density 16.19 kg/m^3
Average kinetic flame temperature 994.17 K
Heat flux density due to metal combustion, $1.04\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $1.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.0058 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $7.52\text{e}-05 \text{ m}$
Pore diameter $9.48\text{e}-06 \text{ m}$
Diffusion flame heat flux density $40,649.76 \text{ W/m}^2$
Diffusion flame height 6,968.45 μm
Heat flux density to surface within skeletal structure $6.38\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $4,944.99 \text{ W/m}^2$
Total heat flux density to surface $6.43\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $6.43\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-1.35\text{e}-05 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 34.41 mm/s

First Conditional Fuel

Surface temperature 828.78 K
Linear burning rate 113.29 mm/s
Mass decomposition rate $187.94 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.14\text{e}+08 \text{ W/m}^2$
Kinetic flame height 2.52 μm
Average kinetic flame density 11.38 kg/m^3
Average kinetic flame temperature 2,267.89 K
Binder sublimation heat flux density $1.14\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.014 W/m^2

Secondary Conditional Fuel

Surface temperature 754.23 K
 Linear burning rate 10.19 mm/s
 Mass decomposition rate $16.91 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Region Outside Skeleton Structure
 Kinetic flame heat flux density to surface $13,835.38 \text{ W/m}^2$
 Kinetic flame height $2,079.98 \mu\text{m}$
 Average kinetic flame density 28.73 kg/m^3
 Average kinetic flame temperature 898.11 K
Skeleton Structure Region
 Kinetic flame heat flux density to surface $2.22\text{e}+06 \text{ W/m}^2$
 Kinetic flame height $21.93 \mu\text{m}$
 Average kinetic flame density 25.86 kg/m^3
 Average kinetic flame temperature 997.61 K
 Heat flux density due to metal combustion, $1.37\text{e}+07 \text{ W/m}^2$
 Effective thermal conductivity $1.42 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $0.0056 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
 Skeleton layer thickness $5.64\text{e}-05 \text{ m}$
 Pore diameter $9.23\text{e}-06 \text{ m}$
 Diffusion flame heat flux density $85,111.89 \text{ W/m}^2$
 Diffusion flame height $3,320.07 \mu\text{m}$
 Heat flux density to surface within skeletal structure $8.31\text{e}+06 \text{ W/m}^2$
 Heat flux density to surface outside skeletal structure $6,616.6 \text{ W/m}^2$
 Total heat flux density to surface $8.4\text{e}+06 \text{ W/m}^2$
 Binder sublimation heat flux density $8.4\text{e}+06 \text{ W/m}^2$
 Surface heat balance error $-2.45\text{e}-04 \text{ W/m}^2$

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
 Calculated burning rate 14.22 mm/s

First Conditional Fuel

Surface temperature 751.62 K
 Linear burning rate 11.34 mm/s
 Mass decomposition rate $18.81 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
 Kinetic flame heat flux density to surface $9.27\text{e}+06 \text{ W/m}^2$
 Kinetic flame height $31.88 \mu\text{m}$
 Average kinetic flame density 1.78 kg/m^3
 Average kinetic flame temperature 2,229.31 K
 Binder sublimation heat flux density $9.27\text{e}+06 \text{ W/m}^2$
 Surface heat balance error 0.001 W/m^2

Secondary Conditional Fuel

Surface temperature 729.36 K
 Linear burning rate 7.44 mm/s
 Mass decomposition rate $12.34 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Region Outside Skeleton Structure
 Kinetic flame heat flux density to surface $116,562.38 \text{ W/m}^2$
 Kinetic flame height $1,403.23 \mu\text{m}$
 Average kinetic flame density 2.57 kg/m^3
 Average kinetic flame temperature 1,547.18 K
Skeleton Structure Region
 Kinetic flame heat flux density to surface $2.64\text{e}+06 \text{ W/m}^2$
 Kinetic flame height $32.65 \mu\text{m}$
 Average kinetic flame density 3.42 kg/m^3
 Average kinetic flame temperature 1,160.18 K
 Heat flux density due to metal combustion, $1.71\text{e}+06 \text{ W/m}^2$
 Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity 0.4 W/(m·K)
 Radiative thermal conductivity 1.64e-06 W/(m·K)
 Conductive thermal conductivity balance error -4.092294297874943E-08
 Skeleton layer thickness 1.35e-04 m
 Pore diameter 1.02e-09 m
 Diffusion flame heat flux density 15,641.41 W/m²
 Diffusion flame height 18,774.75 µm
 Heat flux density to surface within skeletal structure 1.64e+06 W/m²
 Heat flux density to surface outside skeletal structure 72,495.47 W/m²
 Total heat flux density to surface 5.67e+06 W/m²
 Binder sublimation heat flux density 5.67e+06 W/m²
 Surface heat balance error -5.6e-04 W/m²

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
 Calculated burning rate 37.87 mm/s

First Conditional Fuel

Surface temperature 857.12 K
 Linear burning rate 62.2 mm/s
 Mass decomposition rate 103.19 kg·s⁻¹·m⁻²
 Kinetic flame heat flux density to surface 6.72e+07 W/m²
 Kinetic flame height 4.24 µm
 Average kinetic flame density 7.05 kg/m³
 Average kinetic flame temperature 2,282.06 K
 Binder sublimation heat flux density 6.72e+07 W/m²
 Surface heat balance error -0.0029 W/m²

Secondary Conditional Fuel

Surface temperature 775.33 K
 Linear burning rate 17.31 mm/s
 Mass decomposition rate 28.71 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 246,665.47 W/m²
 Kinetic flame height 644.47 µm
 Average kinetic flame density 10.25 kg/m³
 Average kinetic flame temperature 1,570.16 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.73e+07 W/m²
 Kinetic flame height 4.71 µm
 Average kinetic flame density 13.6 kg/m³
 Average kinetic flame temperature 1,183.16 K
 Heat flux density due to metal combustion, 4.2e+06 W/m²
 Effective thermal conductivity 0.4 W/(m·K)
 Conductive thermal conductivity 0.4 W/(m·K)
 Radiative thermal conductivity 1.5e-06 W/(m·K)
 Conductive thermal conductivity balance error -4.092294297874943E-08
 Skeleton layer thickness 5.04e-05 m
 Pore diameter 9.3e-10 m
 Diffusion flame heat flux density 126,824.78 W/m²
 Diffusion flame height 2,279.27 µm
 Heat flux density to surface within skeletal structure 4.31e+06 W/m²
 Heat flux density to surface outside skeletal structure 197,259.31 W/m²
 Total heat flux density to surface 1.52e+07 W/m²
 Binder sublimation heat flux density 1.52e+07 W/m²
 Surface heat balance error -2.78e-05 W/m²

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s
 Calculated burning rate 52.95 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 110.85 mm/s
Mass decomposition rate $183.9 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.32\text{e}+08 \text{ W/m}^2$
Kinetic flame height $2.13 \mu\text{m}$
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density $1.32\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.013 W/m^2

Secondary Conditional Fuel

Surface temperature 793.67 K
Linear burning rate 23.59 mm/s
Mass decomposition rate $39.14 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $309,671.68 \text{ W/m}^2$
Kinetic flame height $507.42 \mu\text{m}$
Average kinetic flame density 16.34 kg/m^3
Average kinetic flame temperature 1,579.33 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $3.18\text{e}+07 \text{ W/m}^2$
Kinetic flame height $2.51 \mu\text{m}$
Average kinetic flame density 21.64 kg/m^3
Average kinetic flame temperature 1,192.33 K
Heat flux density due to metal combustion, $5.82\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.45\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $3.51\text{e}-05 \text{ m}$
Pore diameter $8.99\text{e}-10 \text{ m}$

Diffusion flame heat flux density $247,616.94 \text{ W/m}^2$
Diffusion flame height $1,159.99 \mu\text{m}$
Heat flux density to surface within skeletal structure $6.14\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $259,113.32 \text{ W/m}^2$
Total heat flux density to surface $2.18\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $2.18\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0024 W/m^2

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s
Calculated burning rate 5.97 mm/s

First Conditional Fuel

Surface temperature 822.91 K
Linear burning rate 10.94 mm/s
Mass decomposition rate $18.15 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.09\text{e}+07 \text{ W/m}^2$
Kinetic flame height $26.49 \mu\text{m}$
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature 2,264.95 K
Binder sublimation heat flux density $1.09\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0027 W/m^2

Secondary Conditional Fuel

Surface temperature 782.37 K
Linear burning rate 2.7 mm/s
Mass decomposition rate $4.47 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $325,912.73 \text{ W/m}^2$

Kinetic flame height 485.6 μm
Average kinetic flame density 2.52 kg/m^3
Average kinetic flame temperature 1,573.68 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 199,826.53 W/m^2
Kinetic flame height 404.67 μm
Average kinetic flame density 3.34 kg/m^3
Average kinetic flame temperature 1,186.68 K
Heat flux density due to metal combustion, 1.55e+06 W/m^2
Effective thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 8.95e-08 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.35e-04 m
Pore diameter 5.56e-11 m
Diffusion flame heat flux density 42,376.54 W/m^2
Diffusion flame height 6,804.78 μm
Heat flux density to surface within skeletal structure 451,845.61 W/m^2
Heat flux density to surface outside skeletal structure 241,634.5 W/m^2
Total heat flux density to surface 2.41e+06 W/m^2
Binder sublimation heat flux density 2.41e+06 W/m^2
Surface heat balance error 1.08e-04 W/m^2

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s
Calculated burning rate 13.63 mm/s

First Conditional Fuel

Surface temperature 879.22 K
Linear burning rate 61.8 mm/s
Mass decomposition rate 102.52 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 7.01e+07 W/m^2
Kinetic flame height 4.03 μm
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,293.11 K
Binder sublimation heat flux density 7.01e+07 W/m^2
Surface heat balance error 0.0042 W/m^2

Secondary Conditional Fuel

Surface temperature 803.99 K
Linear burning rate 5.79 mm/s
Mass decomposition rate 9.61 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 741,774.55 W/m^2
Kinetic flame height 210.44 μm
Average kinetic flame density 10.16 kg/m^3
Average kinetic flame temperature 1,584.5 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 105,500.15 W/m^2
Kinetic flame height 745.98 μm
Average kinetic flame density 13.44 kg/m^3
Average kinetic flame temperature 1,197.5 K
Heat flux density due to metal combustion, 3.62e+06 W/m^2
Effective thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 8.23e-08 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5.53e-05 m
Pore diameter 5.11e-11 m
Diffusion flame heat flux density 375,281.52 W/m^2
Diffusion flame height 762.63 μm

Heat flux density to surface within skeletal structure 698,274.41 W/m²
Heat flux density to surface outside skeletal structure 602,647.94 W/m²
Total heat flux density to surface 5.49e+06 W/m²
Binder sublimation heat flux density 5.49e+06 W/m²
Surface heat balance error -1.2e-04 W/m²

At pressure 6.5 MPa

Experimental burning rate 18 mm/s
Calculated burning rate 18.02 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 110.85 mm/s
Mass decomposition rate 183.9 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.32e+08 W/m²
Kinetic flame height 2.13 μm
Average kinetic flame density 11.2 kg/m³
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 1.32e+08 W/m²
Surface heat balance error 0.011 W/m²

Secondary Conditional Fuel

Surface temperature 811.86 K
Linear burning rate 7.57 mm/s
Mass decomposition rate 12.56 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 968,345.02 W/m²
Kinetic flame height 160.39 μm
Average kinetic flame density 16.24 kg/m³
Average kinetic flame temperature 1,588.43 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 84,394.14 W/m²
Kinetic flame height 923.21 μm
Average kinetic flame density 21.47 kg/m³
Average kinetic flame temperature 1,201.43 K
Heat flux density due to metal combustion, 4.87e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 7.99e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.05e-05 m
Pore diameter 4.96e-11 m

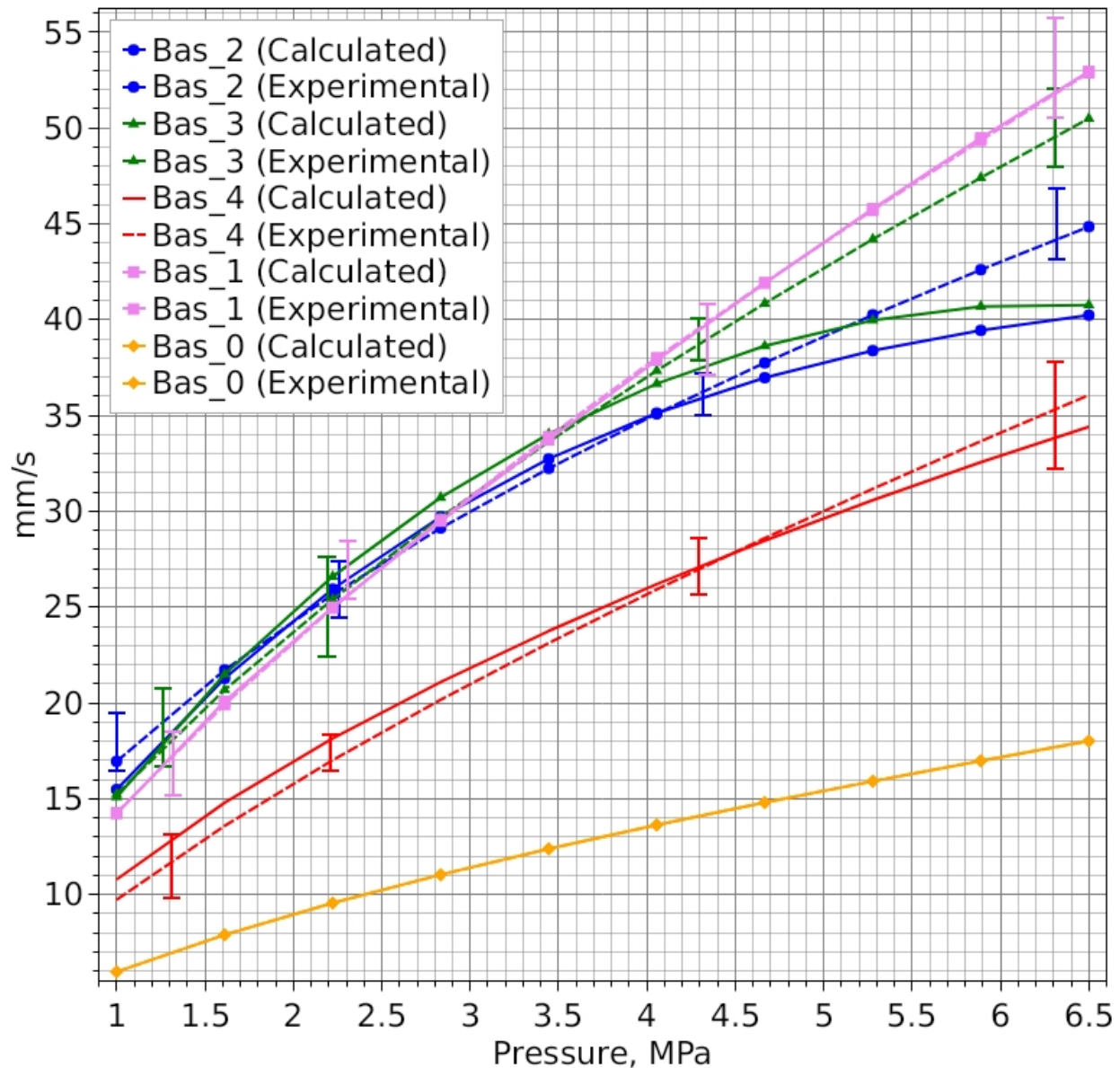
Diffusion flame heat flux density 766,577.68 W/m²
Diffusion flame height 372.32 μm
Heat flux density to surface within skeletal structure 624,318.66 W/m²
Heat flux density to surface outside skeletal structure 846,253.9 W/m²
Total heat flux density to surface 7.33e+06 W/m²
Binder sublimation heat flux density 7.33e+06 W/m²
Surface heat balance error 1.12e-04 W/m²

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	15.48 mm/s		15.09 mm/s		10.79 mm/s		14.22 mm/s		5.97 mm/s	
Difference	1.47 mm/s		0.15 mm/s		-1.06 mm/s		0.048 mm/s		-0.0046 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.3 mm/s	8.31 mm/s	11.3 mm/s	11.3 mm/s	11.3 mm/s	3.77 mm/s	11.34 mm/s	7.44 mm/s	10.94 mm/s	2.7 mm/s
Surface Temperature	757.15 K	748.53 K	757.15 K	757.15 K	757.15 K	727.19 K	751.62 K	729.36 K	822.91 K	782.37 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	9.4e+06 W/m ²		9.4e+06 W/m ²		9.4e+06 W/m ²		9.27e+06 W/m ²		1.09e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	2.27e+06 W/m ²		2.64e+07 W/m ²		871,348.21 W/m ²		2.64e+06 W/m ²		199,826.53 W/m ²	
Kinetic Flame Height (Skeleton)	37.06 μm		7.07 μm		58.97 μm		32.65 μm		404.67 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	104,815.24 W/m ²		77,268.29 W/m ²		4,532.47 W/m ²		116,562.38 W/m ²		325,912.73 W/m ²	
Kinetic Flame Height (Out-Skeleton)	1,542.21 μm		2,080.87 μm		6,945.66 μm		1,403.23 μm		485.6 μm	
Diffusion Flame Heat Flux Density	13,605.98 W/m ²		26,382.67 W/m ²		4,508.11 W/m ²		15,641.41 W/m ²		42,376.54 W/m ²	
Diffusion Flame Height	20,986.91 μm		11,055.95 μm		63,281.75 μm		18,774.75 μm		6,804.78 μm	
Metal Combustion Heat Flux Density	3.23e+06 W/m ²		3.39e+06 W/m ²		4.51e+06 W/m ²		1.71e+06 W/m ²		1.55e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	35.12 mm/s		36.66 mm/s		26.2 mm/s		37.87 mm/s		13.63 mm/s	
Difference	-0.028 mm/s		0.68 mm/s		-0.28 mm/s		0.14 mm/s		-0.0071 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	63.32 mm/s	15.88 mm/s	63.32 mm/s	19.24 mm/s	63.32 mm/s	7.97 mm/s	62.2 mm/s	17.31 mm/s	61.8 mm/s	5.79 mm/s
Surface Temperature	809.45 K	766.93 K	809.45 K	772.56 K	809.45 K	747.34 K	857.12 K	775.33 K	879.22 K	803.99 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	6.09e+07 W/m ²		6.09e+07 W/m ²		6.09e+07 W/m ²		6.72e+07 W/m ²		7.01e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	4.98e+06 W/m ²		6.37e+07 W/m ²		1.75e+06 W/m ²		1.73e+07 W/m ²		105,500.15 W/m ²	
Kinetic Flame Height (Skeleton)	16.53 μm		2.9 μm		28.26 μm		4.71 μm		745.98 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	268,278.25 W/m ²		221,759.83 W/m ²		10,410.06 W/m ²		246,665.47 W/m ²		741,774.55 W/m ²	
Kinetic Flame Height (Out-Skeleton)	595.68 μm		718.09 μm		2,830.5 μm		644.47 μm		210.44 μm	
Diffusion Flame Heat Flux Density	135,646.4 W/m ²		295,480.18 W/m ²		40,649.76 W/m ²		126,824.78 W/m ²		375,281.52 W/m ²	
Diffusion Flame Height	2,091.52 μm		981.94 μm		6,968.45 μm		2,279.27 μm		762.63 μm	
Metal Combustion Heat Flux Density	6.62e+06 W/m ²		6.02e+06 W/m ²		1.04e+07 W/m ²		4.2e+06 W/m ²		3.62e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	40.24 mm/s		40.76 mm/s		34.41 mm/s		52.95 mm/s		18.02 mm/s	
Difference	4.61 mm/s		9.74 mm/s		1.66 mm/s		-0.068 mm/s		-0.017 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	113.29 mm/s	17.52 mm/s	113.29 mm/s	19.69 mm/s	113.29 mm/s	10.19 mm/s	110.85 mm/s	23.59 mm/s	110.85 mm/s	7.57 mm/s
Surface Temperature	828.78 K	769.8 K	828.78 K	773.24 K	828.78 K	754.23 K	900 K	793.67 K	900 K	811.86 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.14e+08 W/m²		1.14e+08 W/m²		1.14e+08 W/m²		1.32e+08 W/m²		1.32e+08 W/m²	
Kinetic Flame Heat Flux Density (Skeleton)	7.28e+06 W/m²		1e+08 W/m²		2.22e+06 W/m²		3.18e+07 W/m²		84,394.14 W/m²	
Kinetic Flame Height (Skeleton)	11.28 μm		1.85 μm		21.93 μm		2.51 μm		923.21 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	414,837.37 W/m²		369,472.69 W/m²		13,835.38 W/m²		309,671.68 W/m²		968,345.02 W/m²	
Kinetic Flame Height (Out-Skeleton)	384.54 μm		430.82 μm		2,079.98 μm		507.42 μm		160.39 μm	
Diffusion Flame Heat Flux Density	329,058 W/m²		773,103.48 W/m²		85,111.89 W/m²		247,616.94 W/m²		766,577.68 W/m²	
Diffusion Flame Height	861.31 μm		375.21 μm		3,320.07 μm		1,159.99 μm		372.32 μm	
Metal Combustion Heat Flux Density	7.38e+06 W/m²		6.17e+06 W/m²		1.37e+07 W/m²		5.82e+06 W/m²		4.87e+06 W/m²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341E-05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

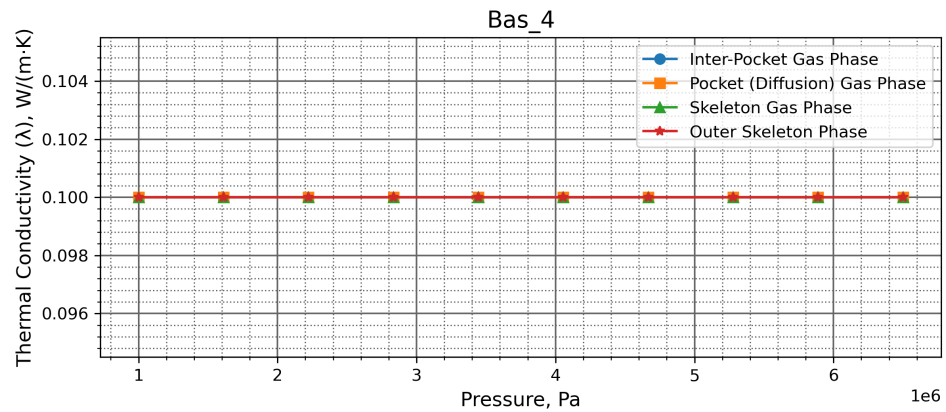
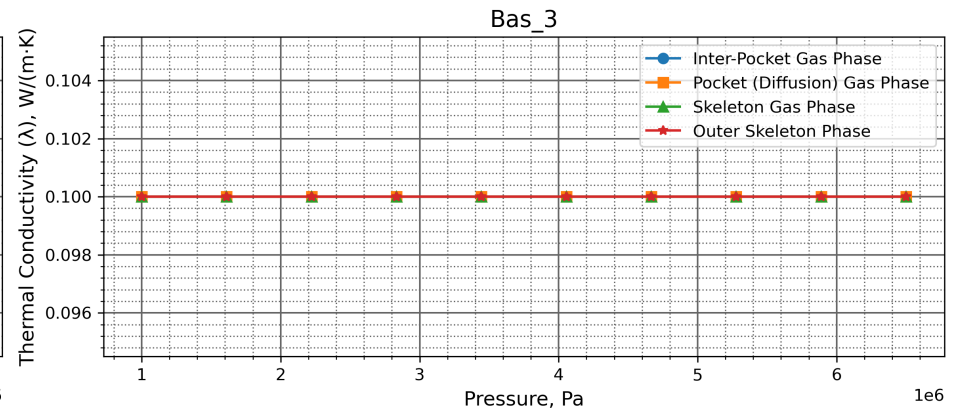
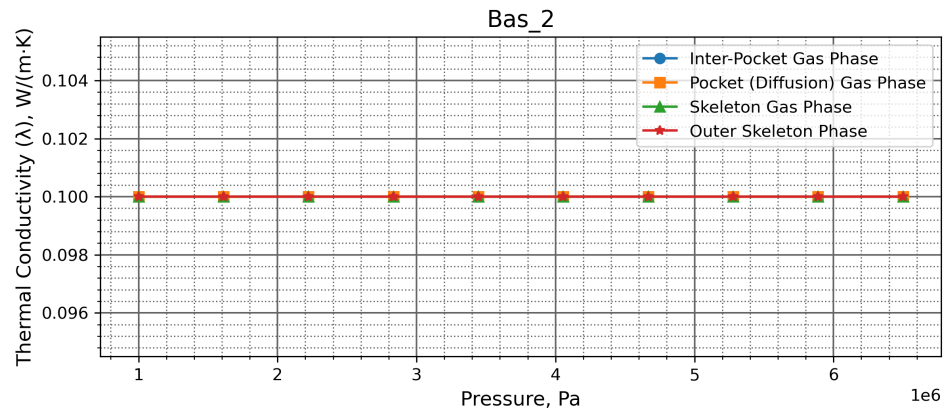
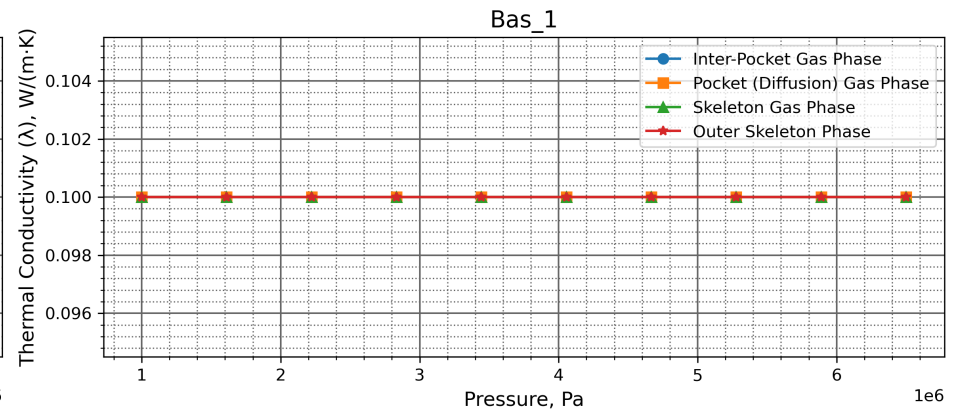
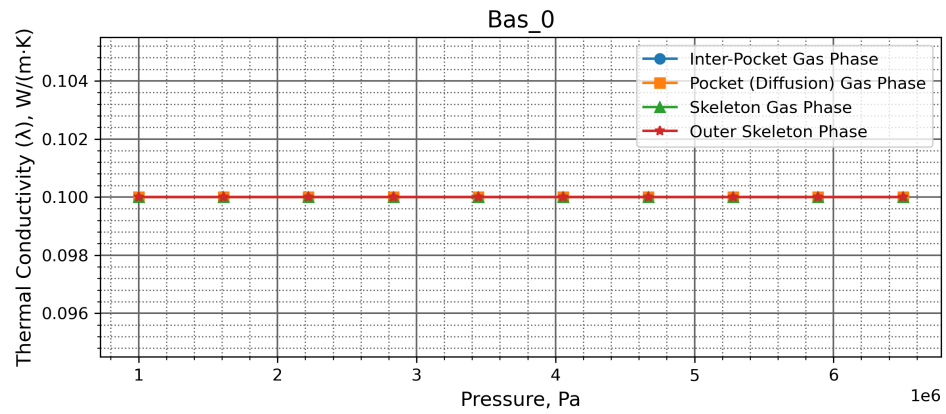
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

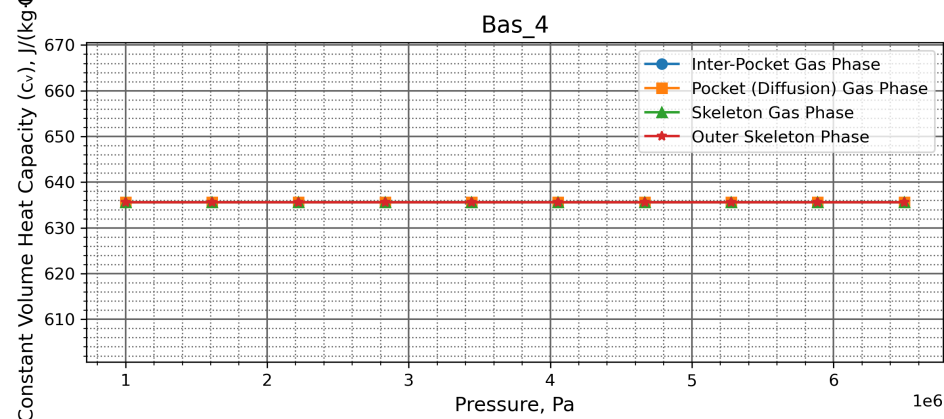
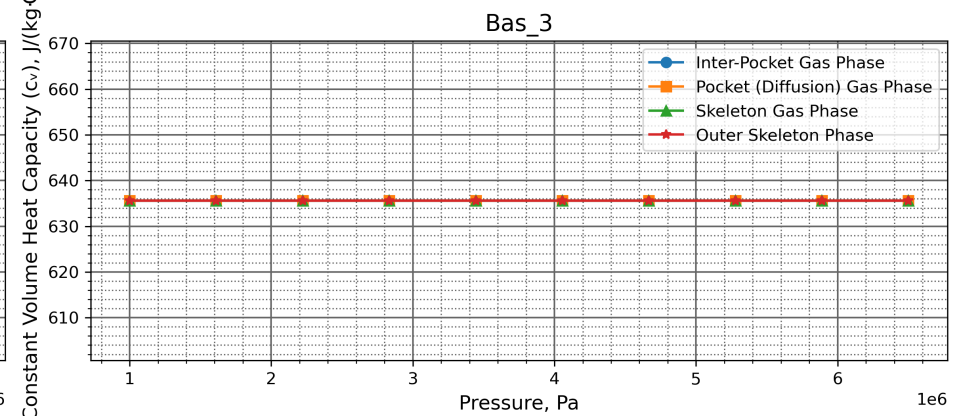
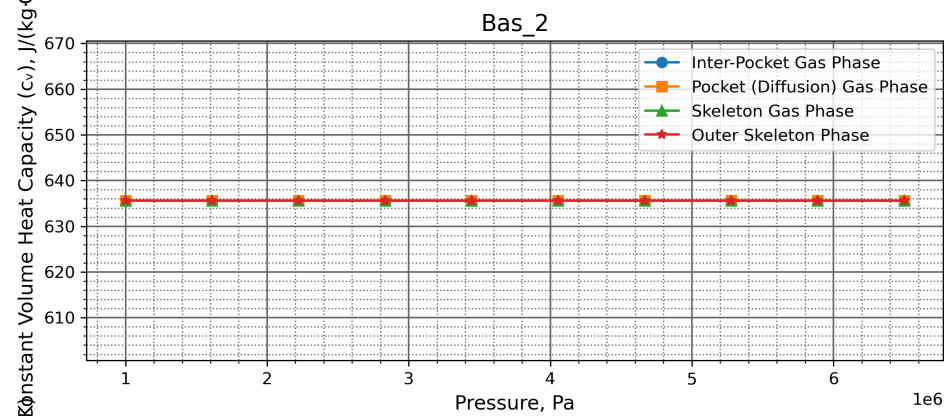
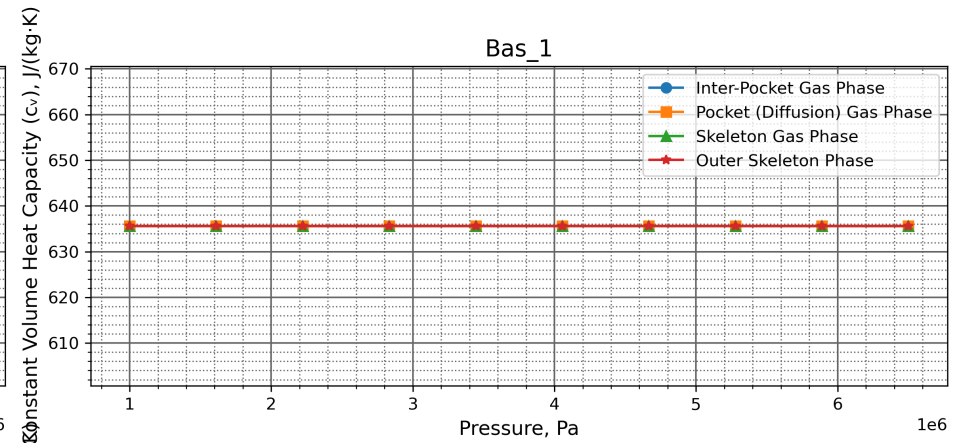
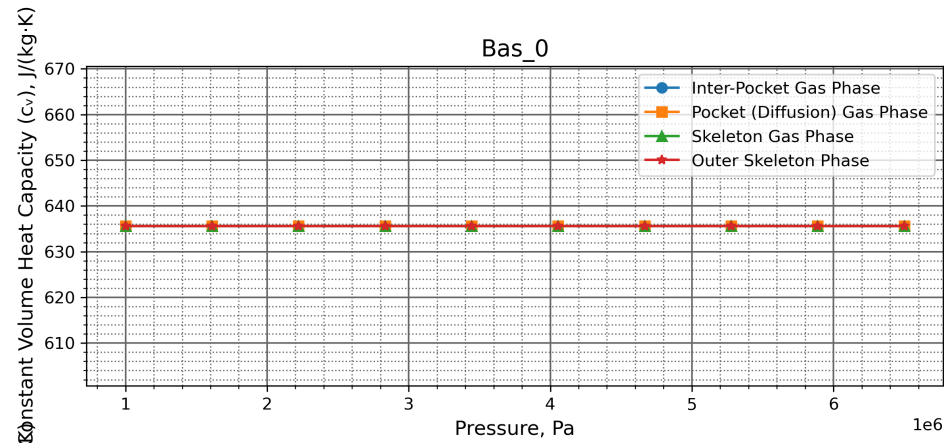
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

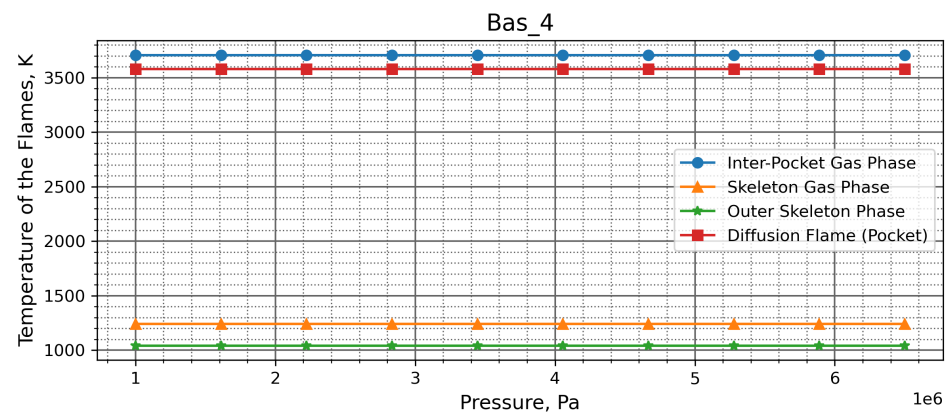
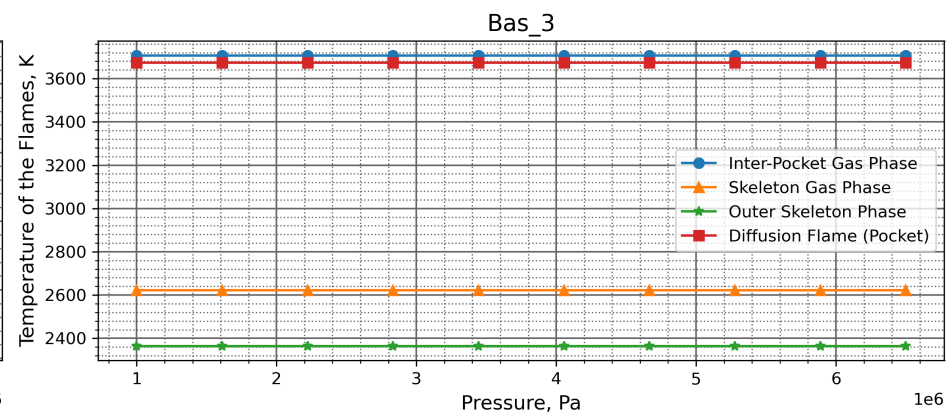
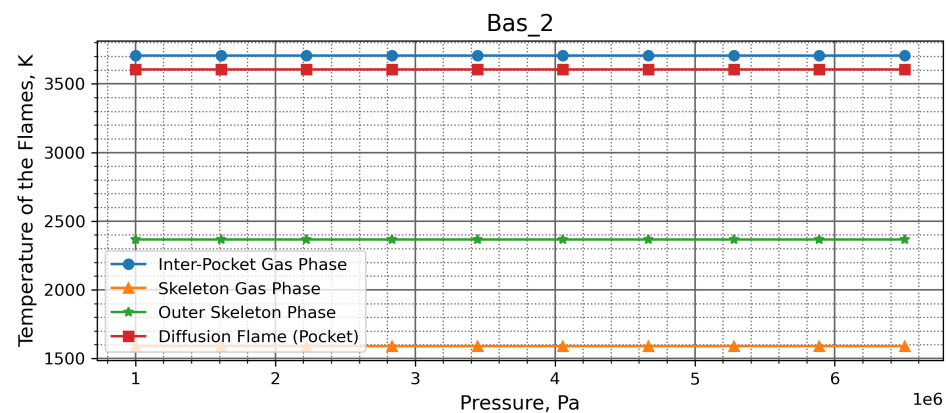
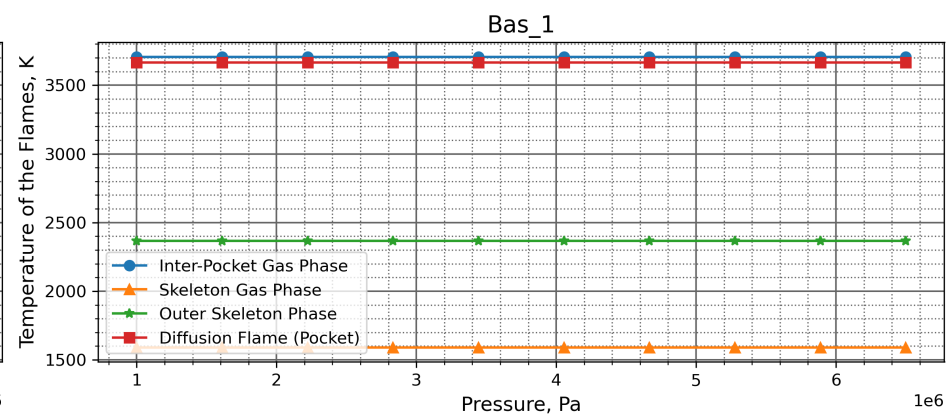
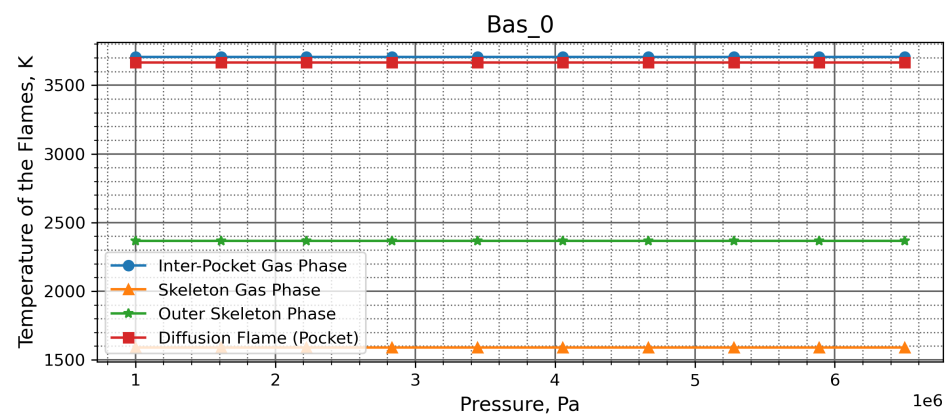
Thermal Conductivity (λ), W/(m·K)



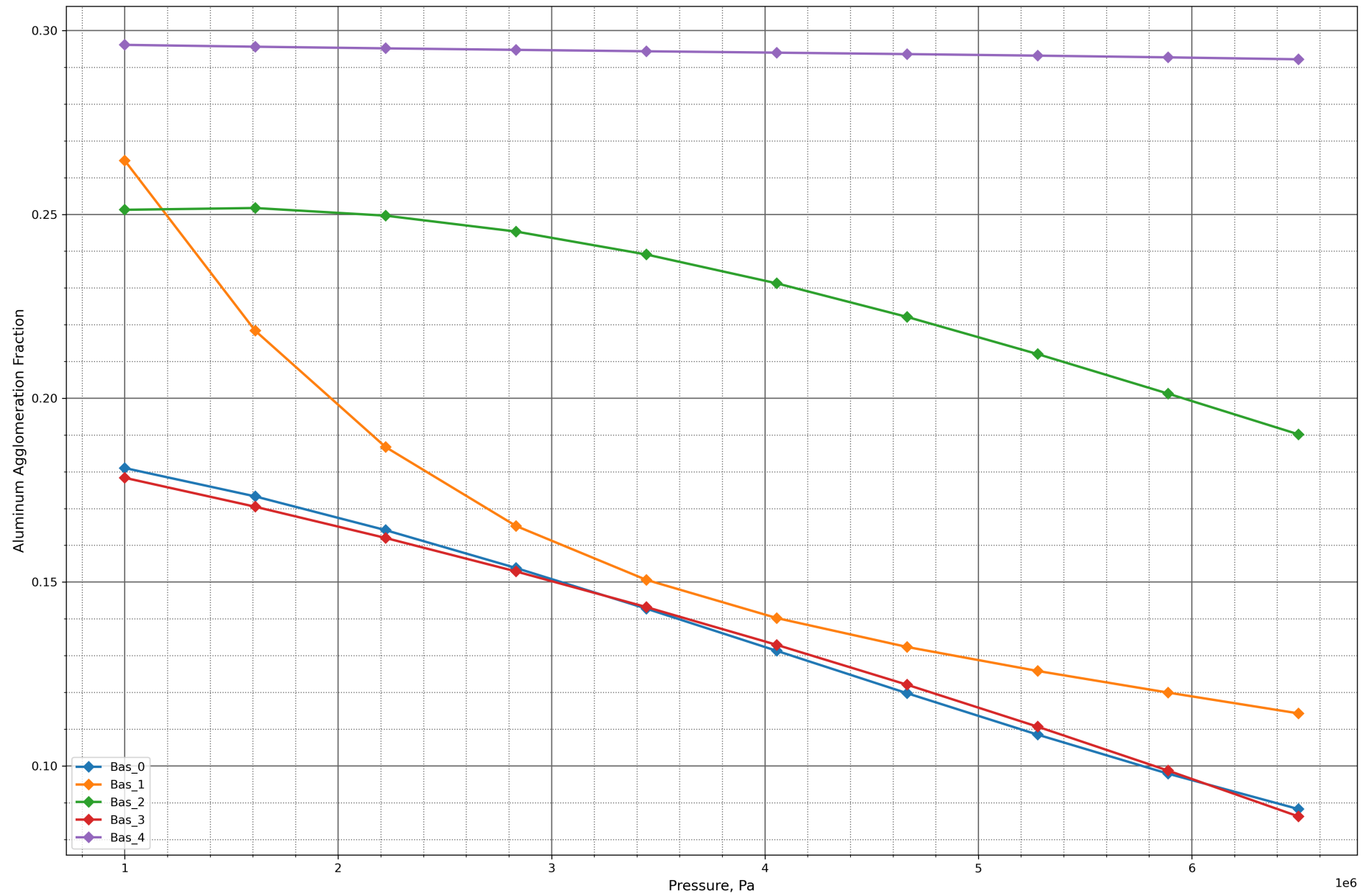
Constant Volume Heat Capacity (c_v), J/(kg·K)



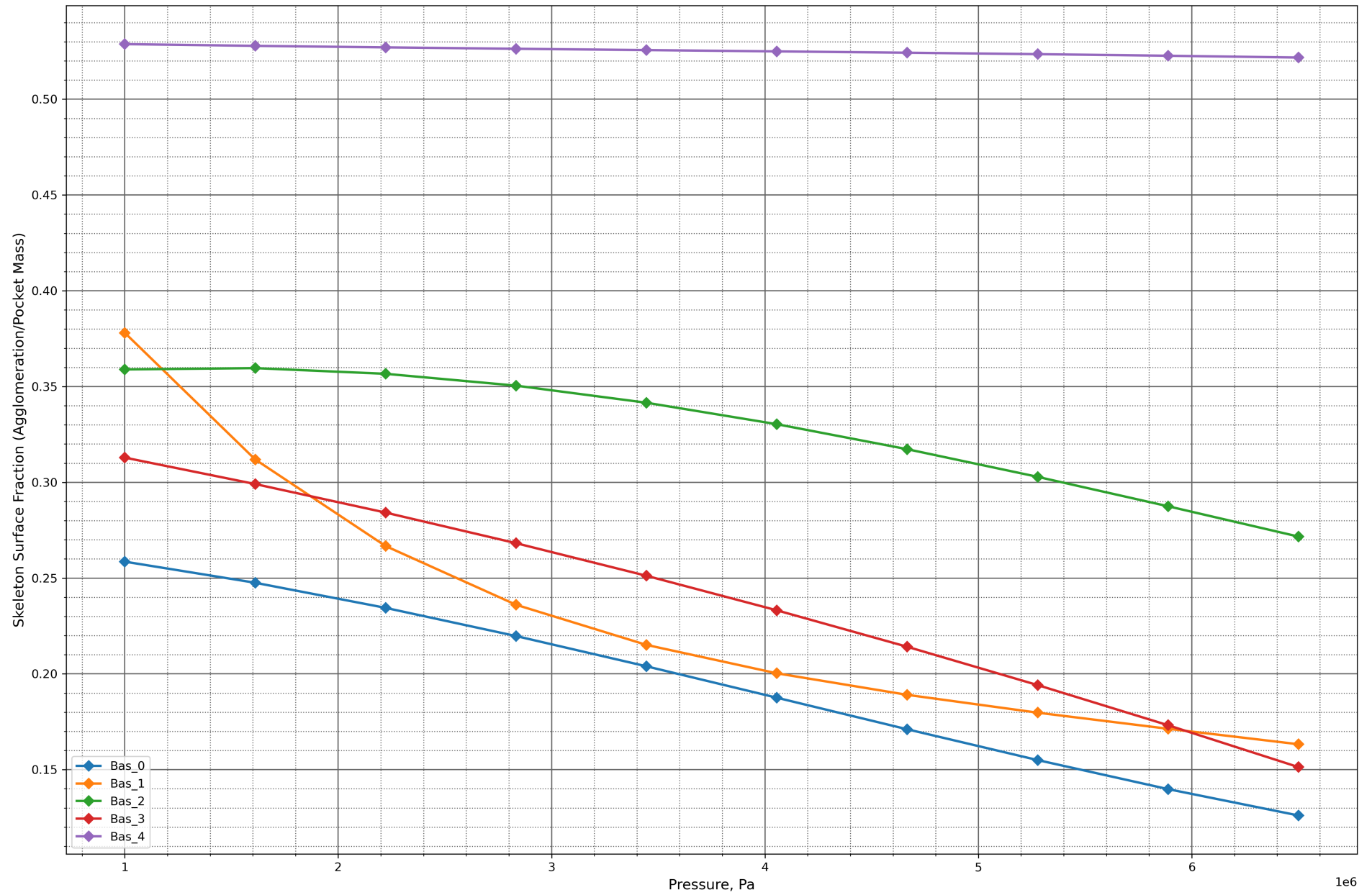
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.124 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.9 (.NET 10.0.9)
Architecture: X64 / X64

Total Execution Time

Total execution time: 09:01:43.455

Garbage Collection Information

Total Memory: 7743400 bytes
GC Pause Time Percentage: 0.10%
Generation 0 Collections: 27977
Generation 1 Collections: 2469
Generation 2 Collections: 275

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 15271975
Maximum execution time: 15.7612 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.9830970208372093 ms
Execution time standard deviation: 0.20061416395474752 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 15090862
Maximum execution time: 19.383 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.3446625979747686 ms
Execution time standard deviation: 0.07429432747478784 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 12681397
Maximum execution time: 15.2857 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34401941865709545 ms
Execution time standard deviation: 0.07654214068359544 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 15171520
Maximum execution time: 12.8292 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9673634579725193 ms
Execution time standard deviation: 0.20236438320913547 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 14992599
Maximum execution time: 11.6266 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33912976372544007 ms
Execution time standard deviation: 0.07565687805920045 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 12566617
Maximum execution time: 7.3889 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.3391642219540962 ms
Execution time standard deviation: 0.07751382682428211 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 16.2879 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9808404603427806 ms
Execution time standard deviation: 0.19510396392213858 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 14996439
Maximum execution time: 11.422 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34399115925457835 ms
Execution time standard deviation: 0.07346800255176164 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 12571558
Maximum execution time: 7.2556 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34425367660870737 ms
Execution time standard deviation: 0.0750741724183553 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 10.4727 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9982734914562944 ms
Execution time standard deviation: 0.1900598528382887 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 14992127
Maximum execution time: 7.5971 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.35059201321468286 ms
Execution time standard deviation: 0.071153742197945 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 12567854
Maximum execution time: 7.1619 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3510470818407317 ms
Execution time standard deviation: 0.07246985583930599 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 12.1269 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9775076452194892 ms
Execution time standard deviation: 0.19323285769990278 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 14997867
Maximum execution time: 8.2535 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3418888596492214 ms
Execution time standard deviation: 0.07227255318567452 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 12584218

Maximum execution time: 15.4838 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34287701550470895 ms

Execution time standard deviation: 0.07434735375539261 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 15171520

Maximum execution time: 10.8029 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9943913179825079 ms

Execution time standard deviation: 0.1962617872320816 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 15000694

Maximum execution time: 9.1636 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34892246134075805 ms

Execution time standard deviation: 0.07285181709307349 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 12576545

Maximum execution time: 8.9468 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34906902920472926 ms

Execution time standard deviation: 0.07472592961348543 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 15171520

Maximum execution time: 15.0468 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.973228588625177 ms

Execution time standard deviation: 0.2043095772643185 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 14988904

Maximum execution time: 14.2158 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3408266161154947 ms

Execution time standard deviation: 0.07598326774136296 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 12556587

Maximum execution time: 58.441 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34154716493417814 ms

Execution time standard deviation: 0.07859279259304516 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 15171520

Maximum execution time: 11.1179 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9794708298839114 ms

Execution time standard deviation: 0.19511332610125223 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 15004801

Maximum execution time: 6.8385 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3436530587776373 ms
Execution time standard deviation: 0.07341881298078849 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 12600269
Maximum execution time: 6.6384 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3438442136433638 ms
Execution time standard deviation: 0.07544283634115384 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 15171520
Maximum execution time: 11.6223 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9861000911443997 ms
Execution time standard deviation: 0.19816959177406623 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 14994930
Maximum execution time: 11.4274 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34626324645064566 ms
Execution time standard deviation: 0.07345597746185281 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 12564123
Maximum execution time: 6.9077 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3459838069239137 ms
Execution time standard deviation: 0.07503865742082119 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 15171520
Maximum execution time: 9.625 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9936911755313905 ms
Execution time standard deviation: 0.19772391891273988 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 14994972
Maximum execution time: 7.442 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3482994000789176 ms
Execution time standard deviation: 0.07360121560585842 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 12573532
Maximum execution time: 7.1744 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3487569402297357 ms
Execution time standard deviation: 0.0744442866300387 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 15171520
Maximum execution time: 17.356 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9792959877059825 ms

Execution time standard deviation: 0.19863256697333212 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 14989250

Maximum execution time: 14.5625 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3432188211150969 ms

Execution time standard deviation: 0.07334413020166367 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 12548336

Maximum execution time: 14.0379 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34306592842272393 ms

Execution time standard deviation: 0.07537984634536317 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 15171520

Maximum execution time: 9.0137 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9973336508076538 ms

Execution time standard deviation: 0.186395423372972 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 14990526

Maximum execution time: 11.5989 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3497384251626699 ms

Execution time standard deviation: 0.06941522782324115 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 12582393

Maximum execution time: 8.9022 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3499948509397175 ms

Execution time standard deviation: 0.07229916243335412 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 15171520

Maximum execution time: 17.6559 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9778136799608192 ms

Execution time standard deviation: 0.1966917096942425 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 14996981

Maximum execution time: 9.8593 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34248688996800075 ms

Execution time standard deviation: 0.07327387874994284 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 12574701

Maximum execution time: 6.6865 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3437886183854886 ms

Execution time standard deviation: 0.0743435718665475 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 15171520
Maximum execution time: 11.7305 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9964806426185554 ms
Execution time standard deviation: 0.21221348532372078 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 14989660
Maximum execution time: 12.5399 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3491377142709701 ms
Execution time standard deviation: 0.07807534117592775 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 12567770
Maximum execution time: 6.8101 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3487002851580046 ms
Execution time standard deviation: 0.0792074295646464 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 15171520
Maximum execution time: 13.8335 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9899801679790905 ms
Execution time standard deviation: 0.20326901120469348 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 15006306
Maximum execution time: 10.9116 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34658632898062464 ms
Execution time standard deviation: 0.07486279438603197 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 12588753
Maximum execution time: 7.7094 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34626792397946077 ms
Execution time standard deviation: 0.07605573681983444 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 15171520
Maximum execution time: 9.2009 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.993150179988583 ms
Execution time standard deviation: 0.19415923273434402 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 14997571
Maximum execution time: 7.1886 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3478337709619531 ms
Execution time standard deviation: 0.07146425117137159 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 12579471
Maximum execution time: 10.7203 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.34783731401748275 ms
Execution time standard deviation: 0.07371760678474043 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 8.913 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0008658280711644 ms
Execution time standard deviation: 0.18651123624289975 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 14996247
Maximum execution time: 21.6585 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.35097680559012007 ms
Execution time standard deviation: 0.07064884011291062 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 12553027
Maximum execution time: 7.0299 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3523236930105189 ms
Execution time standard deviation: 0.07203300872380673 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 18.8885 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9790975718186736 ms
Execution time standard deviation: 0.1999921278027855 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 14999675
Maximum execution time: 8.422 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34298336853963113 ms
Execution time standard deviation: 0.07427419846626881 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 12577608
Maximum execution time: 6.5958 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3434276891838013 ms
Execution time standard deviation: 0.07567484834116261 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 10.4657 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9862096911978585 ms
Execution time standard deviation: 0.19690242516150708 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 14994067
Maximum execution time: 6.3733 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3460806062558125 ms
Execution time standard deviation: 0.07331442218356003 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 12545711
Maximum execution time: 11.248 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34642073508627275 ms
Execution time standard deviation: 0.07480073397599034 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 17.3275 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9774945797653123 ms
Execution time standard deviation: 0.20309955575995478 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 14997593
Maximum execution time: 7.2698 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3424998662852051 ms
Execution time standard deviation: 0.07431678454767517 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 12580215
Maximum execution time: 25.0406 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3435051958889211 ms
Execution time standard deviation: 0.0762881746220689 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 8.8021 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0052077698477508 ms
Execution time standard deviation: 0.1911988422207646 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 14992863
Maximum execution time: 10.6237 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.35289158615674093 ms
Execution time standard deviation: 0.07114159991630352 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 12569003
Maximum execution time: 11.6289 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.35319429705762884 ms
Execution time standard deviation: 0.07252688647075411 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 15.4424 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9782092896624142 ms
Execution time standard deviation: 0.1906607230810452 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 14991844
Maximum execution time: 10.7947 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.34293362669064287 ms
Execution time standard deviation: 0.07172855217161812 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 12553055
Maximum execution time: 6.4348 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3437259810540414 ms
Execution time standard deviation: 0.07316786410574944 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 10.4547 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9855512453399111 ms
Execution time standard deviation: 0.1898217229349356 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 14990912
Maximum execution time: 7.1302 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34622200872769576 ms
Execution time standard deviation: 0.07030117093922945 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 12553800
Maximum execution time: 10.4491 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.346721491580229 ms
Execution time standard deviation: 0.07196307722803738 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 11.0618 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.984613885879609 ms
Execution time standard deviation: 0.1925562664473672 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 15001731
Maximum execution time: 21.4093 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3450692045737213 ms
Execution time standard deviation: 0.07201919928084549 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 12593218
Maximum execution time: 9.9082 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34555321089493934 ms
Execution time standard deviation: 0.0733831702748969 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 15171520
Maximum execution time: 23.9842 ms
Minimum execution time: 0.0002 ms
Mean execution time: 1.0057361039367345 ms
Execution time standard deviation: 0.20882803876654296 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 14995764

Maximum execution time: 11.3725 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3523526302427859 ms

Execution time standard deviation: 0.0767056953745564 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 12559313

Maximum execution time: 14.3725 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3522665426444975 ms

Execution time standard deviation: 0.07765711135921827 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 15171520

Maximum execution time: 13.1319 ms

Minimum execution time: 0.0002 ms

Mean execution time: 1.00066357209409 ms

Execution time standard deviation: 0.192085780339875 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 14997893

Maximum execution time: 14.1108 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.35040591153036965 ms

Execution time standard deviation: 0.0716641958641459 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 12599385

Maximum execution time: 6.447 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.350962702600216 ms

Execution time standard deviation: 0.07344723067001696 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 15171520

Maximum execution time: 12.7102 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9910038992598625 ms

Execution time standard deviation: 0.18978438751871757 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 14992845

Maximum execution time: 12.4336 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3470854476318401 ms

Execution time standard deviation: 0.07089590674675111 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 12575604

Maximum execution time: 8.4059 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34732264204567703 ms

Execution time standard deviation: 0.07287123113734621 ms